EGG SEGMENTATION & VIDEO TRACKING

Computer Vision Intern Task — Technical Report

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Public GitHub repository
https://github.com/basavarajasuti69/egg-segmentation

Output video — two-ID egg tracking
Open the final tracking video in Google Drive
https://drive.google.com/file/d/1vIVKLtKufem7F6pb_bMC0VSyC-Na3iEo/view?usp=sharing

This report documents a workflow for generating reference egg masks, evaluating classical image segmentation, and tracking two eggs in a video. Metrics and file counts below are based on the recorded project results.

1. Project Overview

The objective was to segment eggs in still images, compare classical segmentation approaches against reference masks, and apply a tracking pipeline to a video. The work was organized into four tasks: reference-mask generation, development/evaluation split and experiment logging, quantitative and qualitative evaluation, and video tracking.

The final classical approach selected on the development set was a hybrid segmentation method with shape and area validation and a GrabCut fallback. For video, the final output uses two track labels (Egg 1 and Egg 2) to match the two visible eggs. The tracker combines filtered detections with centroid-based matching. The tracker is constrained to two slots; this prevents label proliferation but does not prove perfect identity preservation through occlusions.

2. Task 1 — Reference Masks

A pretrained segmentation workflow was used to generate reference masks for 110 dataset images. Saved outputs include instance/mask and overlay artifacts. The recorded rough image distribution was 92 images with one egg, 7 with two eggs, and 11 with three eggs.

The masks serve as approximate reference annotations for evaluating the classical method. Automatically generated masks can include boundary errors or missed/merged regions, so Task 3 scores represent agreement with these generated references, not necessarily human-verified ground truth.

3. Task 2 — Classical Segmentation Experiments

The dataset was divided into a development set and a held-out evaluation set using a 4:1 split: 88 development images and 22 evaluation images. Experiments were recorded in an experiment log, and the selected configuration was frozen after development-set comparison.

Trial	Mean IoU	Median IoU	Mean Dice	Images IoU < 0.5
1	0.3428	0.2393	0.4607	64
2	0.3545	0.2470	0.4562	65
3	0.4949	0.5099	0.5924	43
4	0.1778	0.0000	0.2014	72
5	0.4957	0.4570	0.6055	49
6	0.5718	0.6320	0.6719	37
5 (rerun)	0.6010	0.6674	0.6941	33
5 (rerun)	0.5911	0.6363	0.6859	35
7 (selected)	**0.6405**	**0.8057**	**0.7022**	**26**

Selected method: Trial 7 — hybrid segmentation with shape and area validation, using GrabCut as a fallback. Development-set results: mean IoU 0.6405, median IoU 0.8057, mean Dice 0.7022; 26 of 88 images had IoU below 0.5.

4. Task 3 — Held-Out Evaluation

The frozen selected algorithm was evaluated on the 22-image held-out evaluation split. No further tuning was performed on this split.

Metric	Held-out result
Evaluation images	22
Mean IoU	0.53199

Median IoU	0.58182
Mean Dice	0.60393
Images with IoU < 0.5	10
Images with IoU ≥ 0.5	12

The evaluation scores are lower than the development-set results, indicating that the selected method does not generalize equally well to all held-out images. Difficult cases include images where background, lighting, shadows, or egg boundaries challenge classical segmentation. The submission package includes a CSV of per-image metrics and visualizations of selected worst cases.

5. Task 4 — Video Tracking

The supplied YouTube clip was processed as a 60-second video at 24 frames per second (1,440 frames; 1920 × 1080). The final tracking output uses two labels, Egg 1 and Egg 2.

Tracking summary	Recorded result
Video duration / frame rate	60 seconds / 24 FPS
Total frames	1,440
Detection records	1,229
Frames containing detections	763
Track ID 1 observations	630
Track ID 2 observations	599

Final output video: Google Drive — egg_tracking_two_ids.mp4

Limitations: hands and partial occlusions can interfere with detections, and fixed two-slot tracking does not guarantee perfect identity continuity in every frame. Some frames may contain no detections. The result should be interpreted as a practical demonstration rather than a fully robust multi-object tracking system.

6. Reproducibility and Submission Contents

The project repository is intended to contain the source notebook/code, requirements, experiment log, evaluation CSV/JSON outputs, this report, and a link to the final output video. The dataset and large video should be hosted separately where appropriate rather than committed directly to GitHub.

Repository: https://github.com/basavarajasuti69/egg-segmentation

Final video: Google Drive share link

Environment/setup note: install dependencies from requirements.txt, open the project notebook in Google Colab or a compatible Python environment, and update any data/video paths to match the local or Drive folder layout. Exact reproducibility depends on matching dependency versions and providing the required dataset and input video.

7. AI Tool Disclosure

AI assistance was used to help organize the workflow, troubleshoot implementation issues, interpret evaluation outputs, and prepare documentation. The reported metrics and counts in this report are taken from the recorded project outputs. The author should review the final code, results, and disclosure to ensure they accurately reflect the work completed and the task's rules.

Before final submission, confirm the GitHub repository is public, the notebook/source code and supporting result files are uploaded, and the Google Drive video is accessible to anyone with the link.